

Preprocessing Steps and Business Questions Analysis Report

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Project Title:	India’s Agricultural Crop Production Analysis

1. Introduction

This document presents the data preparation process, business requirements, and analytical questions addressed in the **India Agricultural Crop Production Analysis (1997–2021)** project. The objective of this project is to analyze agricultural production trends across various states, crops, and seasons using Tableau dashboards and visualizations.

2. Dataset Description

Parameter	Description
Dataset Name	India Agricultural Crop Production Dataset
Duration	1997–2021
Domain	Agriculture
Tool Used	Tableau
Data Attributes	State, District, Crop, Season, Area, Production, Yield, Year
Data Source	Agricultural Production Dataset

3. Data Preprocessing Steps

Since the dataset was already structured and clean, only validation and preparation activities were performed.

Step No.	Preprocessing Activity	Description
1	Data Import	Imported the dataset into Tableau
2	Data Validation	Verified column names and data types
3	Missing Value Check	Checked for null values and inconsistencies
4	Duplicate Verification	Confirmed absence of duplicate records
5	Data Categorization	Grouped data by State, Crop, Season, and Year
6	Measure Aggregation	Applied SUM aggregation to production and yield values
7	Filter Creation	Created filters for State, Crop, Season, and Year

4. Business Questions

Which states contribute the highest agricultural production in India?

Visualization Used:

- Horizontal Bar Chart
- State-wise Production Dashboard

Insight:

Punjab, Uttar Pradesh, Maharashtra, and Andhra Pradesh emerged as major agricultural contributors.

Business Question 2

Which crops have the highest production over the years?

Visualization Used:

- Bar Chart
- Crop Performance Dashboard

Insight:

Rice, Wheat, Sugarcane, and Cotton showed the highest production levels.

Business Question 3

How has agricultural production changed from 1997 to 2021?

Visualization Used:

- Line Chart
- Trend Analysis Dashboard

Insight:

Agricultural production exhibited an overall increasing trend with fluctuations due to seasonal and climatic factors.

Business Question 4

Which agricultural season contributes the most to crop production?

Visualization Used:

- Pie Chart / Donut Chart

Insight:

Kharif and Rabi seasons account for the majority of agricultural production.

Business Question 5

What is the relationship between cultivated area and crop production?

Visualization Used:

- Scatter Plot / Comparative Chart

Insight:

Higher cultivation area generally leads to increased production, although crop yield efficiency varies by state and crop type.

Business Question 6

Which states have demonstrated consistent agricultural growth?

Visualization Used:

- Line Chart
- State-wise Trend Dashboard

Insight:

Several states exhibited stable agricultural growth due to improved farming practices and technological advancements.

5. KPI Analysis

The following Key Performance Indicators (KPIs) were developed:

KPI	Description
Total Production	Total agricultural production
Total Yield	Overall crop yield
Number of States	States included in analysis
Number of Crops	Total crop categories
Number of Seasons	Agricultural seasons analyzed
Year Range	1997–2021

6. Dashboard Summary

Dashboard	Description	Visualizations
Dashboard 1	National Agricultural Overview	KPI Cards, Bar Chart, Pie Chart
Dashboard 2	State-wise Agricultural Analysis	Maps, Bar Charts, Trend Charts
Dashboard 3	Crop and Seasonal Analysis	Line Charts, Donut Charts, Comparisons

7. Story Summary

The Tableau story consists of **6 story points** demonstrating:

1. Agricultural production overview
2. State-wise production analysis
3. Crop-wise performance
4. Seasonal contribution analysis
5. Production trend analysis
6. Final business insights and recommendations

8. Conclusion

The Tableau-based Agricultural Crop Production Analysis successfully identified trends, regional variations, seasonal impacts, and crop performance patterns across India from 1997 to 2021. Interactive dashboards and story points enabled efficient decision-making and comprehensive agricultural insights.